Steele 2020 PDAC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

13/13 samples (≥117 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→5.15 of 13); both lobes (Ductal-classical 6,782 / Stress-UPR 1,092) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

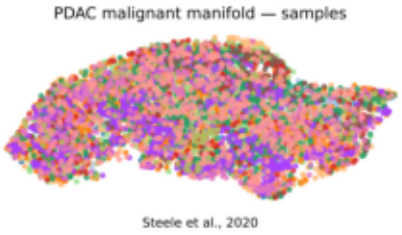
Single dominant pSom→Crest axis: pSom high over the classical-ductal body declines along pseudotime (ρ=−0.70); Crest anti-correlated (r=−0.69), rising to a terminal readout (ρ=+0.74; crest-high decile mean pt=0.87). Gast weakly structured (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.44) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

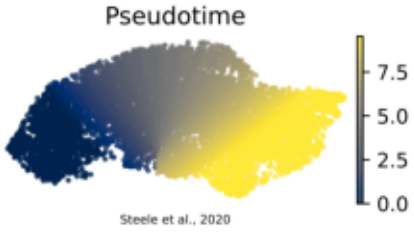
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

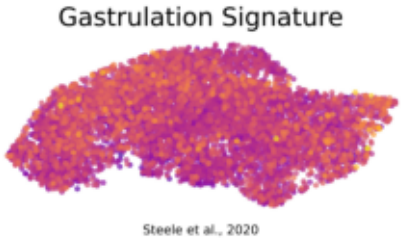
Manifold — samples



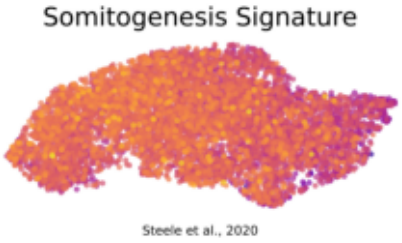
Pseudotime



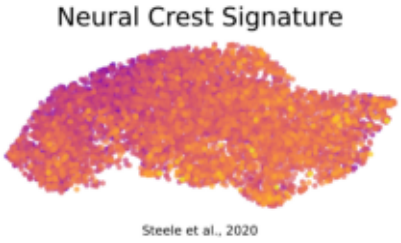
Gastrulation



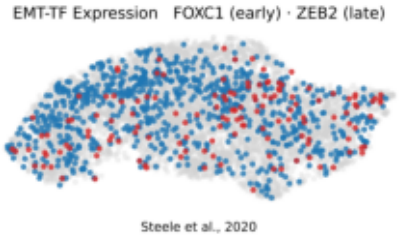
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

PDAC (Steele et al., 2020), 10x — 16 patients / 48,570 cells (master scope); malignant only: 7,874 cells across 13 malignant-bearing samples (all clear the ≥100 floor) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Steele's PDAC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (classical-ductal body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both transcriptional lobes (orthogonal to cell cycle; module–cyc |ρ|≤0.08), so the whole manifold is kept (Bischoff/Sun/Lee no-scoping precedent). Unlike Zhou, cell cycle does not define the lobes (PDAC ~94% non-cycling) and pseudotime is uncorrelated with cycling (ρ=+0.02), so no proliferation caveat applies. The raw module means (Gast −0.04 / pSom +0.08 / Crest −0.05) nearly match the failed Peng 2019 PDAC run, but the geography differs: Peng showed three mutually-exclusive territories with no terminal readout (NO-GO), whereas Steele shows a coherent terminal-crest gradient (GO) — the gate turns on geometry, not means. EMT-TFs FOXC1 (early/root) → ZEB2 (late/terminus); canonical SNAI1/ZEB1 near the detection floor; EMT-TF end-separation weak (Δpt=0.12).

Parameter choices

Min cells/sample	≥100 floor (13/13; gap degenerate)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Ductal-cl. 6,782 / Stress-UPR 1,092